II-MAP: LOGBOOK

Kramp Thomas

Universiteit van Antwerpen Toegepaste Ingenieurswetenschappen
(Elektronica-ICT)

Date	Action
05/01/2026	– Setup docs: Paper, Portfolio, Logbook & Reports – Add LLC review to Portfolio
07/01/2026	– Create LLC schematic
09/01/2026	– Create LLC PCB
11/01/2026	– Continue working on LLC PCB
19/01/2026	– Add new LLC review to portfolio (no more JLCPCB) – Disassemble TruggySense v1 – Add temperature sensor review to portfolio – Create and finish temperature sensor schematic and PCB
20/01/2026	– Figured out the external ADC usage (conflicting documentation)
21/01/2026	– Made changes to the temperature sensor schematic and PCB to adhere to school components – Added the defense of ADC usage in the portfolio. – Started on figuring out the power supply (missing documentation).
23/01/2026	– Finished figuring out the power supply (missing documentation). – Made a schematic of missing power supply documentation.
25/01/2026	– Added the defense for the HLC, Motion sensor, Wheel actuator and Ampere meter in the portfolio. – Started on the defense for the Position sensor, RC controller and wifi in the portfolio. – Worked on the defense of the supply and DC/DC converter in the portfolio. – Added power use expectation table. – Added I^{2}C address use table.
27/01/2026	– Added the defense for the Motion sensor, Position sensor in the portfolio. – Added reasoning of portfolio v1 to portfolio v2. – Added Kiwi, Conrad and Antratek to each stock table. – Updated power use expectation table. – Updated I^{2}C address use table.
01/02/2026	– Completed the defense for the LLC, Long range communication and Current sensor in the portfolio.
02/02/2026	– Added defense for the DC/DC converter to the portfolio. – Added Otronic to the stock tables. – Made it so only two of the following web-shops show: Kiwi, Conrad, Antratek and Otronic. – Changed the long-range communication defense based on feedback.
03/02/2026	– Added Shunt resistor calculations to the current meter defense. – Create a schematic and PCB for the Power Supply PCB.

06/02/2026	– Added KiCAD guides. – Cleaned up and sorted old .stl files and their FreeCAD/OpenSCAD file. – Created an overview of all external libraries and their references. – Created a defense for and made changes to the IO protection circuits.
08/02/2026	– Recreated the PCB layout for the nRF24L01 and PIM448 modules. – Recreated the PCB for the INA219 and LM2596 modules. – Continued on the IO protection defense.
09/02/2026	– Edited the IO protection circuits based on feedback. – Created a CSV for the low-pass filter/voltage divider circuit. – Created a schematic for the position module.
16/02/2026	– Created a PCB for the position module. – Started on the logic PCB schematic.
18/02/2026	– Continued on the logic PCB schematic.
19/02/2026	– Finished the logic PCB schematic. – Started on the logic PCB design.
20/02/2026	– Continued on the logic PCB design.
24/02/2026	– Made the powerpoint for the first presentation.
25/02/2026	– Gave the first progress report presentation.
26/02/2026	– Continued on the logic PCB design.
27/02/2026	– Finished the logic PCB design.
01/03/2026	– Generated all the gerber, drill and BOM files. – Listed all parts with stocking urls for order.
02/03/2026	– Installed platformIO. – Figured out how to create a Teensy project via PlatformIO. – Documented process.
05/03/2026	– Created test code for Teensy 4.1, I2C interface, INA219. – Created simple libraries for I2C interface and INA219.
06/03/2026	– Created test code for SD-Card. – Tried to make test code for nRF24L01, FS-X6B and ACS758; but couldn't due to insufficient resources. Found compatible libraries that are ready for testing. – Tested the FlySky remotes and removed any faults.
06/03/2026	– Figured out how to use FreeCAD and started its guide.
09/03/2026	– Cleaned up the old 3D-design sketches. – Edited the old 3D-design sketches.
10/03/2026	– Edited the old 3D-design sketches. – Created a .dxf file for the new lasercut baseplate.

15/03/2026
- Edited the old 3D-design sketches.
- Started with ROS2 & micro-ROS implementation (not functioning).
- Started working with Nvidia Jetson Nano (not functioning).

16/03/2026
- Printed all the finished 3D-prints.
- Continued to work on the Nvidia Jetson Nano (not functioning).

17/03/2026
- Got the Nvidia Jetson Nano to work.
- Continued work on the ROS2 & micro-ROS implementation (not functioning).

18/03/2026
- Lasercut the base plate.
- Made changes to the new 3D-design sketches so they fit the baseplate.

20/03/2026
- Got a base implementation of ROS2 & micro-ROS working between the Teensy 4.1 and my personal laptop.

21/03/2026
- Updated AI-usage documentation.

24/03/2026
- Documented the complete KiCAD section.

26/03/2026
- Created a diagram of the power layout.

27/03/2026
- Fetched the components and PCBs.
- Assembled the power board and temperature sensor PCBs.

28/03/2026
- Documented the complete FreeCAD and Metalcut sections.
- Updated the ROS integration of the Software chapter for both the Teensy and the Jetson.

29/03/2026
- Created test code for LM75 and PIM448.
- Created simple libraries for LM75 and PIM448.

30/03/2026
- Assembled the first temperature sensor board.
- Assembled the power board.
- Started assembly of the logic board.
- Tested the temperature sensor.

31/03/2026
- Finished assembly of the logic board.
- Tested the power distribution of the logic board.

01/04/2026
- Assembled the GNSS/GPS module.
- Assembled the remaining temperature sensor boards.
- Created a PWM test.
- Tested the INA219, LM2596, PIM448.

03/04/2026
- Created a servo script.
- Tested the remaining temperature sensors and servo.
- Tested the power cycle between power board, TMDC and logic board.
- Continued documentation of design.
- Documented PCB faults.

13/04/2026
- Tested the voltage followers.
- Found a fault with the buffer circuit.
- Created a test for USB. –¿ Blew up my PC.

15/04/2026	– Reinstalled everything on new PC. – Tested old functioning sketches, to make sure board was still functioning.
16/04/2026	– Created and tested the FS-X6B sketch. – Created and tested the Hall-sensor sketch.
17/04/2026	– Created and tested the high-current sensing. – Tested the self-made GPS (not functioning).
18/04/2026	– Found the motor ECS startup sequence. – Fixed the motor sketch. – Created a working drive sketch.

20/04/2026	– Created the main sketch. – Implemented the remote control in the main sketch. – Implemented the steering servo in the main sketch.
21/04/2026	– Implemented the motor control in the main sketch. – Implemented the voltage reading in the main sketch.
22/04/2026	– Implemented a global debugging serial in the main sketch. – Implemented the encoder in the main sketch. – Tried to implement the high current reading in the main sketch.
23/04/2026	– Implemented the high current reading in the main sketch. – Implemented the low current reading in the main sketch. – Implemented the temperature sensor in the main sketch.
24/04/2026	– Implemented the indicator led in the main sketch. – Implemented the cooling fan in the main sketch. – Implemented the motion sensor in the main sketch.
25/04/2026	– Tried to create a GPS sketch.
26/04/2026	– Tried to make a long range communication sketch.

27/04/2026	– Made a long range communication TX and RX sketch. – Made a long range communication RX module. – Implemented the long range TX communication in the main sketch.
28/04/2026	– Implemented the SD-logging into the main sketch. – Started to implement the schedular into the main sketch.
29/04/2026	– Continued to implement the schedular into the main sketch.
30/04/2026	– Finished to implement the schedular into the main sketch. – Debugged the main sketch for any mistakes.
01/05/2026	– Debugged the main sketch for any mistakes. – Made a functioning GPS sketch. – Tried to implement micro-ROS into the main sketch.
02/05/2026	– Implemented the GPS into the main sketch. – Tested the complete system and found bugs. – Removed the bugs.

03/05/2026	– Implemented .csv and .json logging via the SD-Card (besides .bin). – Tested the complete system and found bugs. – Removed the bugs. – Started on the academic paper.
04/05/2026	– Continued working on the academic paper.
05/05/2026	– Continued working on the academic paper.
06/05/2026	– Continued working on the academic paper.
07/05/2026	– Continued working on the academic paper. – Went on the RC Racing Fun track to test the system and gather data. – Replaced the broken Servo.
08/05/2026	– Continued working on the academic paper. – Went on the RC Racing Fun track to test the system and gather data.
09/05/2026	– Continued working on the academic paper. – Fixed the GPS baud rate, by programming the module inside the firmware. – Started on working on the next PCB. – Went on the RC Racing Fun track to test the system and gather data.
10/05/2026	– Continued working on the academic paper. – Attempted to fix the wheel encoder bug (untested on the track).
11/05/2026	– Fixed the wheel encoder bug. – Fixed one of the broken servos. – Worked on the documentation. – Started on the finalized PCB designs (temperature sensor and supply board).
12/05/2026	– Worked on the documentation. – Continued on the finalized PCB designs (logic board). – Went on the RC Racing Fun track to test the system and gather data, but couldn't due to faulty GPS antenna.
13/05/2026	– Worked on the documentation. – Continued on the finalized PCB designs (logic board).
14/05/2026	– Worked on the documentation. – Finished the finalized PCB designs (logic board). – Started adding documentation to the code.
15/05/2026	– Generalized the motor control code (now includes both ESC and SERVO). – Finished adding documentation to the code. – Went on the RC Racing Fun track to test the system and gather data, but couldn't due to the state of the track and weather conditions. – Fixed problems caused by driving on a less than ideal track.
16/05/2026	– Added more functionality to the SD-logging. – Tested the complete system (in my front yard).
17/05/2026	– Went on the RC Racing Fun track to test the system and gather data, got base data that proves functionality, but couldn't continue due to the state of the track and weather conditions. – Fixed problems caused by driving on a less than ideal track.

18/05/2026	– Worked on the academic paper. – Worked on the documentation.
19/05/2026	– Worked on the academic paper. – Worked on the documentation.
20/05/2026	– Worked on the documentation.
21/05/2026	– Re-read and corrected the academic paper. – Worked on the documentation.
22/05/2026	– Went on the RC Racing Fun track to test the system and gather data. – Implemented the new data into and finished the academic paper.
23/05/2026	– Re-read and corrected the academic paper. – Worked on the documentation.
24/05/2026	– Re-read and corrected the academic paper. – Finished on the documentation.